

Ink in Motion: Re-drawing Beijing's Eight Sights Through Immersive Dance

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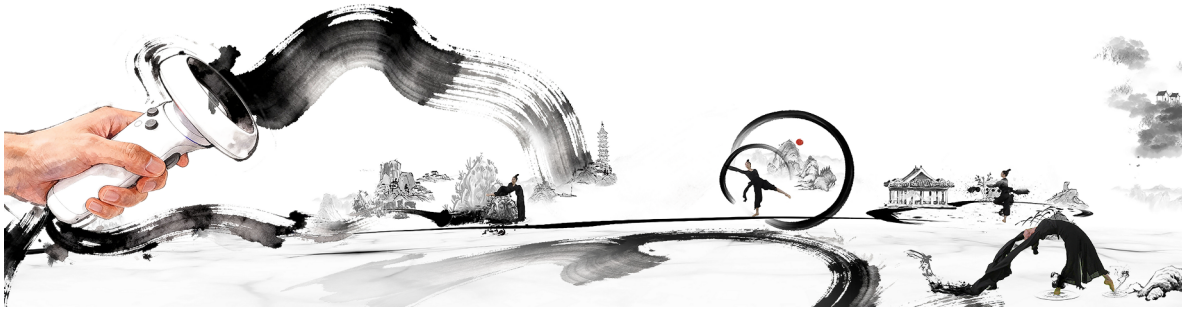


Figure 1: Conceptual Rendering of Ink in Motion.

ABSTRACT

Ink in Motion is an interactive virtual reality dance experience that reimagines classical Eight Sights of Yanjing (ancient Beijing) through embodied ink choreography. Presented as a hybrid panoramic film, the work transforms the dancer's motion—and the viewer's gestures—into a shared medium of calligraphic ink. Flowing sleeves generate landscapes in real time, while hand-tracked interaction allows participants to draw ink directly into the virtual space, forming a responsive duet between performer and audience.

By combining motion-captured dance, physically based cloth simulation, and gesture-driven ink rendering, the piece explores motion as medium, where cultural memory emerges through bodily movement rather than static representation. *Ink in Motion* re-frames traditional landscape imagery as a living, co-created process, inviting viewers to re-experience heritage through embodied participation. The work demonstrates how immersive XR can transform spectatorship into authorship, offering a poetic model for re-drawing cultural memory across past, present, and future.

Index Terms: Virtual Reality; Motion Capture; Immersive Dance; Cultural Heritage; Gesture-Based Interaction; Cloth Simulation.

1 DESCRIPTION OF ARTWORK

Ink in Motion is an immersive VR dance experience that reimagines Beijing's classical landscape through the material and metaphorical qualities of ink. The 360° journey through the Eight Sights of Yanjing follows a dancer whose flowing sleeves generate the scenery itself—each gesture releasing digital ink that forms rivers, bridges, and mountains, merging choreography with brush-stroke.

As the performance evolves, ink becomes the connective tissue between body, history, and space. The dancer paints, dissolves,

and reconstitutes the landscapes within a living ink world, suggesting that cultural memory is not static but continuously re-written through embodied motion.

For interaction, the viewer is invited to join the dance. Through real-time hand tracking, their gestures generate streams of ink that ripple through the virtual space and merge with the dancer's trails. When the viewer extends a hand, the ink flows outward, responding to the velocity, curvature, and orientation of the movement. Slow, circular gestures evoke mist and clouds; sharp, linear gestures release splashes that coalesce into brush-like structures. The dancer's sleeve motion and the user's ink strokes resonate in synchrony, composing a duet between human performer and digital participant [4].

The work thus transforms the traditional act of “watching a dance” into an embodied act of “co-creating choreography.” The viewer is no longer an external observer but a co-author of the visual narrative, engaging directly in the re-drawing of Beijing's cultural topography through motion and imagination [3, 2].

2 EMOTIONAL CONNECTION AND TIME RESHAPING

The work connects audiences with tradition across three intertwined temporal layers—historical, contemporary, and speculative—where ink shifts from fixing meaning to flowing with gesture, visualising cultural memory as continuity in transformation.

- **Historical:** The Eight Sights of Yanjing—icons of classical poetry and painting—embody ancient Beijing's cultural identity.
- **Contemporary:** A live dancer performs within a volumetric capture setting, expressing through movement the embodied continuation of traditional aesthetics in modern media.
- **Speculative:** The digital ink environment, generated in real time through participants' gestures, envisions how future observers may reinterpret the heritage through computational creativity.

3 IMPLEMENTATION AND TECHNICAL EXECUTION

Ink in Motion is a hybrid panoramic film + interactive VR piece for Meta Quest 3, combining volumetric dance, cloth simulation, and

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real-time gesture-driven ink rendering to bridge cinematic movement and embodied participation.

3.1 Panoramic Capture and Digital Performance

The foundational layer of the piece is a 360° dance film, captured with a multi-camera panoramic rig in a controlled studio environment (Fig. 2). The dancer's choreography was designed to visualise the gestural logic of calligraphy: sleeve rotations correspond to brush curves, and body spirals mirror the radial diffusion of ink.

To preserve the material dynamism of motion, a digital double of the performer was constructed in Maya using a motion-capture dataset drawn from professional dance libraries. The Maya model was animated through the captured motion curves and exported via FBX into Houdini and After Effects pipelines for compositing.



Figure 2: Green-screen studio capture of the dancer.

3.2 Cloth Simulation: “Ink as Fabric”

A central visual motif—the dancer's sleeve transforming into cascades of liquid ink—was achieved through a high-fidelity cloth simulation pipeline [1].

- In Maya, a simplified 3D mesh approximating the dancer's pose and timing was designated as the collision object.
- Using the nCloth system, a virtual fabric with low stiffness and high damping was assigned, allowing folds to “pour” like viscous liquid.
- The simulated fabric was rendered in two passes with ink-wash-style textures featuring dynamic transparency and edge erosion.

This hybrid method allowed motion itself to become the pigment—the body's trajectory continuously paints and erases, embodying the notion that gesture and medium are inseparable.

3.3 Real-Time Interaction and XR Integration

The interactive adaptation extends this cinematic foundation into a real-time VR environment built in Unity using the XR Interaction Toolkit.

- Hand Tracking: Quest 3's native tracking captures skeletal joint transforms for each frame, yielding fine motion vectors.
- Ink Generation: A GPU compute-shader converts these vectors into procedural ink flows governed by Navier–Stokes-style advection, producing realistic diffusion and absorption [5].
- Choreographic Coupling: Pre-recorded dancer motion modulates local turbulence and particle density. When the viewer's ink stream intersects the dancer's virtual sleeve path, the two motion fields blend to form a shared “ink duet.”

3.4 Sound and Immersion

Spatialised audio is rendered through Quest 3's integrated speakers, calibrated for full 3D panning. Original soundscapes combine traditional guqin and xiao tones processed with granular synthesis. Sound parameters respond to dancer velocity and user gesture—slow motions yield overtones, vigorous ones trigger percussive splashes. This audio-kinetic mapping deepens bodily immersion without external hardware.

In this environment, the viewer's gestures extend the dancer's performance, dissolving the historical divide between artist and spectator. Every movement—recorded, simulated, or spontaneous—joins an evolving calligraphic space where body writes and ink remembers.

4 ARTIST INFORMATION

Tianxing Zhou is a PhD Candidate at the School of Communication and Arts, The University of Queensland. Her research tackles the narrative paradox in VR interaction, developing approaches where gameplay and storytelling co-amplify rather than compete. She serves as a reviewer for ACM Multimedia 2025 and a Program Committee Member for AICS 2025. Her award-winning VR work *So Long* has been exhibited at over twenty venues worldwide. Her work also received the Best Sound Design Award at the UNESCO Digital Heritage Contest. She holds a Master's degree in Immersive Storytelling from University College London, where she received the Best Dissertation Award and was named to the Dean's List.

Zihan Chen is a graduate student at the School of Intelligent Image Engineering, Beijing Film Academy. His primary research focuses on AI-enabled optimization throughout the film and television production pipeline, virtual production, and XR-based shooting. His works have been shortlisted for a host of renowned awards, including selections for the Milan Design Week, nominations for the UK Culture and Arts Awards, and finalist positions at the National Digital Design Competition of China.

Xiaoxuan Liu is a graduate student at the Beijing Film Academy Digital Media School, specializing in Game Creation. During her studies, she was awarded the National Scholarship and recognized as an Outstanding Graduate of Beijing, among other honors. Her artistic practice focuses on VR digital art, and she has established a workflow centered on interactive engine technology. Her works have garnered dozens of awards and nominations, including the Outstanding Award at the 32nd College Student Film Festival of the Beijing International Film Festival and the Gold Award at the Busan International Art Festival.

ACKNOWLEDGMENTS

Tianxing Zhou was supported by The University of Queensland through a Graduate School Scholarship

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